Visvesvaraya Technological University, Belagavi

JSS Mahavidyapeetha
JSS Academy of Technical Education, Bengaluru
Department of Information Science and Engineering

JSSATE BENGALURU

Project Presentation on

Early Detection of Chronic Kidney Disease Through Quantum Machine Learning Model

Presented by

Batch No	Name	USN
08	MITHUN GOWDA HS	1JS22IS072
	MURULIDHARA H G	1JS22IS078
	PRADEEP ISHWAR GOUDA	1JS22IS088
	ROHAN M	1JS22IS113

Guide Name: Dr. Naidila Sadashiv
Designation: Professor JSSATEB

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Introduction

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- Chronic Kidney Disease (CKD) is a long-term medical condition characterized by the gradual and irreversible loss of kidney function.
- CKD often progresses silently, showing no early symptoms, which leads to late diagnosis and severe complications.
- According to global health studies, over 115 million people in India are affected by CKD, making it a major public health concern.
- Early detection is critical to prevent progression to end-stage renal disease (ESRD) and reduce mortality.

Problem Statement

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This project focuses on developing a quantum machine learning–based system to enable early detection of Chronic Kidney Disease using clinical data. By employing a Quantum Support Vector Machine, the model improves diagnostic accuracy and supports timely medical intervention through an automated and web-based decision support system.

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Objectives

- To develop a Quantum based model for early detection of Chronic Kidney Disease (CKD).
- To implement and compare Quantum Machine Learning (QML) with other classical ML Algorithms.
- To develop simple users friendly web application to check whether CKD is present or not.
- To evaluate the model using accuracy and related performance metrics.

Software Requirements

Programming Language and Version :- Python , 3.8 or above

Web Framework :- Python Flask (for backend development)

Machine Learning Libraries :- NumPy , Pandas , Scikit-learn …

Data Preprocessing Tools :- KNN Imputer , OneHotEncoder , RobustScaler

Quantum Computing Libraries :- PennyLane (Library) , default.qubit (as a simulator)

Frontend Technologies :- HTML , CSS , JavaScript

Development Environment :- Jupyter Notebook

Browser :- Google Chrome

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Hardware Requirements

Processor :- Intel Core i5 / i7 or AMD Ryzen 5 / Ryzen 7

RAM :- Minimum 8 GB

Storage :- 20 GB free disk space

Operating System :- Mac / Windows 10 / Windows 11

Internet Connection :- Required for installing libraries, tools, and running the web application

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Dataset

- **Obtained from public UCI Machine Learning Repository :- https://archive.ics.uci.edu/dataset/336/chronic+kidney+disease**
- **Total 400 Patient Records with 25 Attributes.**
- **Numerical Attributes :- 11 (ex:- Age , Blood Pressure , Haemoglobin etc)**
- **Categorical Attributes :- 13 (ex:- Hypertension , Diabetes , Anemia etc)**

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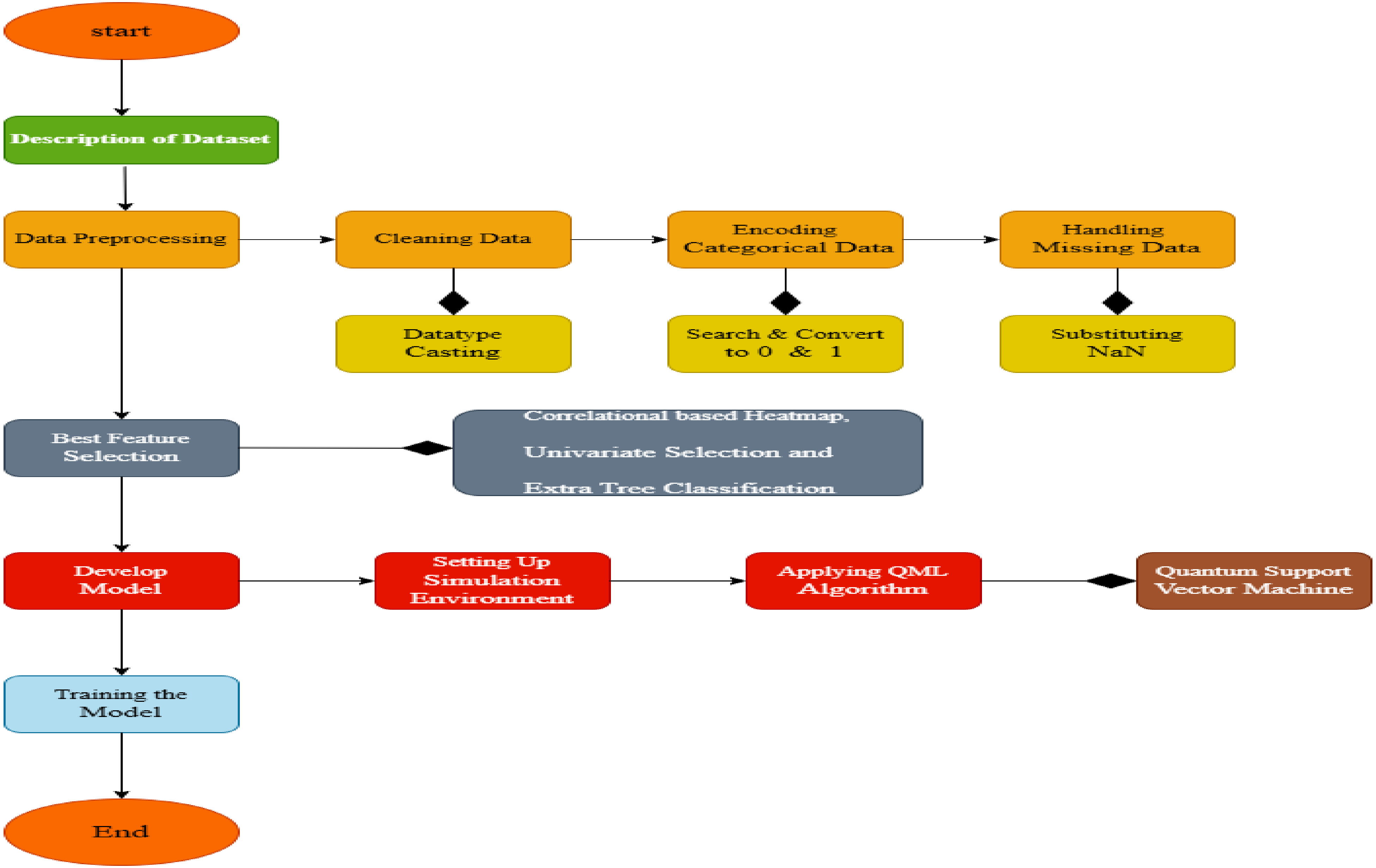
Methodology



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ML Model: QSVM Algorithm

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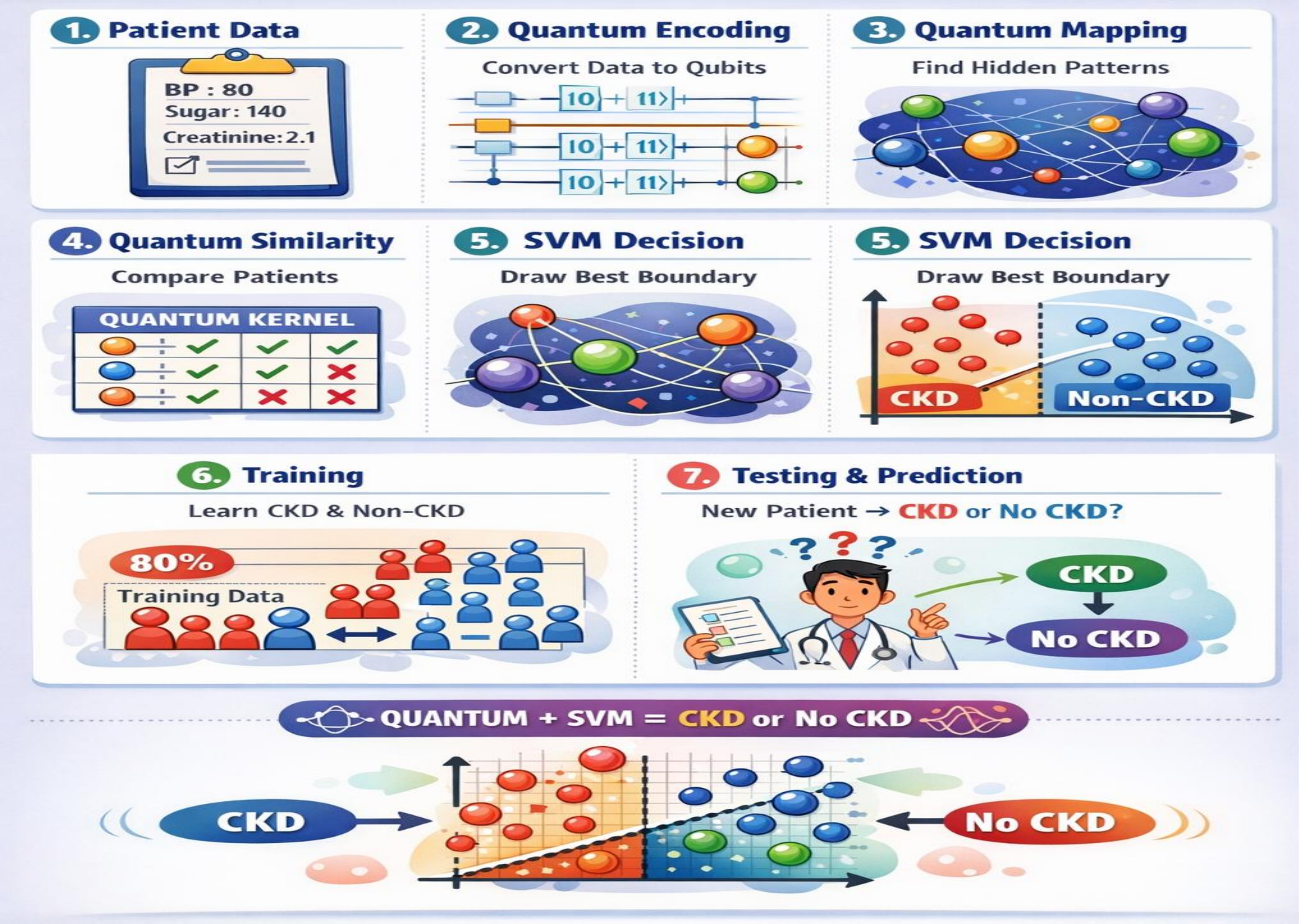

Challenges and Limitations

Dataset Size:
Dataset size is relatively less of having only 400 patient records .

Model Integration:
Integrating the QSVM model into the Flask backend for real-time predictions was challenging.

Computational Constraints:
Quantum model simulation on classical hardware increases processing time.

Hardware Dependency:
Execution of quantum algorithms depends on high-performance systems and simulators.

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Project Outcomes

- Implemented a system for early detection of Chronic Kidney Disease (CKD) using clinical and biochemical patient parameters.
- Performed data preprocessing and anomaly handling, including missing value imputation, normalization, and dimensionality reduction.
- Utilized a Quantum Support Vector Machine (QSVM) integrated with classical machine learning techniques to predict CKD presence at an early stage.
- Achieved high accuracy and sensitivity, enabling reliable identification of CKD cases with minimal false negatives.
- Developed a user-friendly web interface using Python Flask to input patient data and display prediction results with confidence scores.
- Designed a hybrid classical–quantum framework using efficient, scalable, and reproducible machine learning pipelines.

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Conclusion & Future Scope

- Our project uses a Quantum Support Vector Machine (QSVM) to detect Chronic Kidney Disease at an early stage. The model achieves high accuracy and performs better than traditional machine learning methods. The developed web application makes the system easy to use and suitable for future healthcare applications.

- Our goal is to collect a sufficient amount of data in the future so that we can perform the validation efficiently.
- Using Noninvasive measurement does not require a blood sample and reduces the sufferings related to invasive methods such as discomfort, infection risk, pain,

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